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Vision-based geometric finite element model updating for cable suspension bridges

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ABSTRACT

The geometric characteristics of a structure are crucial in determining the accuracy and reliability of finite element (FE) models. However, conventional FE models, which are typically based on theoretical specifications from the design stage, often fail to account for geometric discrepancies introduced during fabrication and construction processes. To address this limitation, this study proposes a methodology that employs computer vision techniques to refine geometric inconsistencies in FE models using point cloud data. The proposed method achieves a closer match between the FE model and the actual target structure by aligning it with point cloud data and refining nodal coordinates. A simulation test was conducted to ensure that the updated FE model accurately represents the geometric characteristics of the target structure, and its practical applicability was validated through an experimental test under actual conditions. This method could further enhance the applicability of FE models in civil engineering, and is expected to contribute to the development of state-of-the-art solutions for computer vision-based geometric FE model updating.

1. Introduction

Safety management of infrastructure is a crucial task which is directly related to public safety. Various approaches are employed to achieve this, with on-site inspections and monitoring technologies being the most commonly employed. On-site inspections offer the advantage of directly assessing the condition of structures through visual observations, equipment-based measurements, and field tests [1–3]. However, they are time-consuming, costly, and often rely on the subjective judgment of the inspector. In contrast, monitoring technologies involve the installation of sensors on structures to collect real-time data, which is then analyzed to assess the structural integrity [4–6]. This approach offers significant advantages in terms of efficiency and objectivity. However, the high initial installation cost and the need for complex data interpretation limit its practical applicability. To overcome these limitations, various data analysis and interpretation techniques have been continuously developed to identify and assess the condition of structures using measurement data [7–11].

One of the most widely used methods for evaluating the condition of structures using measurement data is vibration-based system identification [12–14]. This method analyzes dynamic response data of the structure in the frequency domain to extract dynamic characteristics such as natural frequencies and damping, and uses this information to estimate changes in the structural condition. While this approach is relatively simple to apply and has high practical utility, it faces challenges in obtaining sufficient long-term data for structural changes to become apparent. Additionally, the extracted structural characteristics are sensitive to environmental factors such as temperature, making it difficult to directly assess the condition of the structure.

There is another approach for assessing the structural condition using measurement data, which is damage detection [15–19]. This method evaluates the structural condition by identifying differences in response between damaged and undamaged conditions. It may involve statistical analysis methods, various signal processing techniques, and advanced technologies like artificial intelligence. However, damage detection methods are generally based on simulation data that can consider various damage conditions, which may not fully represent the actual behavior of the structure, limiting the practical applicability of such methods.

In addition to analyzing measurement data, another method for evaluating the condition of structures is the use of finite element (FE) models. FE models numerically analyze the behavior of a structure by

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dividing it into finite elements, reflecting the geometric and material properties as well as boundary conditions of the target structure. However, in real structures, various uncertainties during fabrication and construction processes inevitably introduce discrepancies between the FE model and the actual structure. To address this issue, FE model updating methods have been introduced, which adjust the parameters of the model to minimize the differences between the responses of the FE model and those of the actual target structure [20–24]. These methods primarily use dynamic responses of the structure for updating purposes, with its material properties considered as key variables.

In an FE model, the geometric properties serve as the basis for the application of material properties and boundary conditions. However, since the geometric characteristics of an FE model are typically determined based on theoretical design specifications, such models often fail to account for errors or imperfections arising during the fabrication and construction processes of the actual structure. This issue becomes particularly significant for structures with complex geometric features, such as cable bridges, where such geometric discrepancies can have a substantial influence on the overall behavior of the structure. Therefore, there is a need for the development of technologies that can adjust the geometric characteristics of the FE model to better match those of the actual target structure [25–27].

In this regard, this study proposes a computer vision-based FE model updating method to overcome such geometric discrepancies. The method utilizes computer vision techniques to obtain point cloud data of the target structure, which is then matched with the FE model to accurately adjust its geometric characteristics. The main contribution of this study lies in the integration of computer vision, geometric representation, and FE model updating, offering a novel framework for calibrating the geometric features of the target structure.

The following chapter describes the proposed geometric FE model updating method in detail. Chapter 3 verifies the theoretical validity of the proposed method through simulations. Chapter 4 evaluates the practical applicability of the methodology through experimental tests. Finally, chapter 5 presents the main conclusions of the study and outlines directions for future research.

2. Methodology

Fig. 1 shows a schematic description of the point cloud-based geometric FE model updating method proposed in this study. The method is based on the following assumption regarding the point cloud: The point cloud extracted from the target structure is densely distributed in areas that exhibit the geometric features of the actual structure, while noise is relatively sparse. These characteristics can be further ensured with advancements in point cloud acquisition and data processing technologies. Under this assumption, the proposed method proceeds in three steps: first, the acquisition of point cloud data for the target structure; second, the alignment of the coordinate systems between the point cloud and the theoretical FE model; and third, the geometric updating of the FE model based on the aligned point cloud.

2.1. Point cloud acquisition

There are several methods for obtaining 3D point clouds of a target using imaging devices. These include LiDAR (Light Detection and Ranging) [28–30], which captures 3D points by emitting a laser and measuring the time it takes for the light to reflect; stereo cameras [31–33], which extract depth information by comparing images taken from multiple cameras at different angles; and SfM (Structure from Motion) [34–36], which reconstructs the camera position and 3D structure based on the relative motion between multiple images taken from a single camera at different angles. Each method has its own advantages. LiDAR is advantageous in terms of high accuracy and long-distance measurement, while stereo cameras are relatively affordable and capable of real-time processing. On the other hand, SfM is beneficial because it can efficiently measure complex structures without the need for expensive equipment. In this study, SfM was used to collect point cloud data of the target structure, leveraging the advantages of this technology.

SfM is a computer vision technique that reconstructs a 3D point cloud structure of a target object by utilizing images taken from various angles, as shown in Fig. 2. This method detects feature points from each image, calculates the camera position and orientation through feature matching, and then derives the 3D information of the target object, i.e., the point cloud, based on optimization techniques.

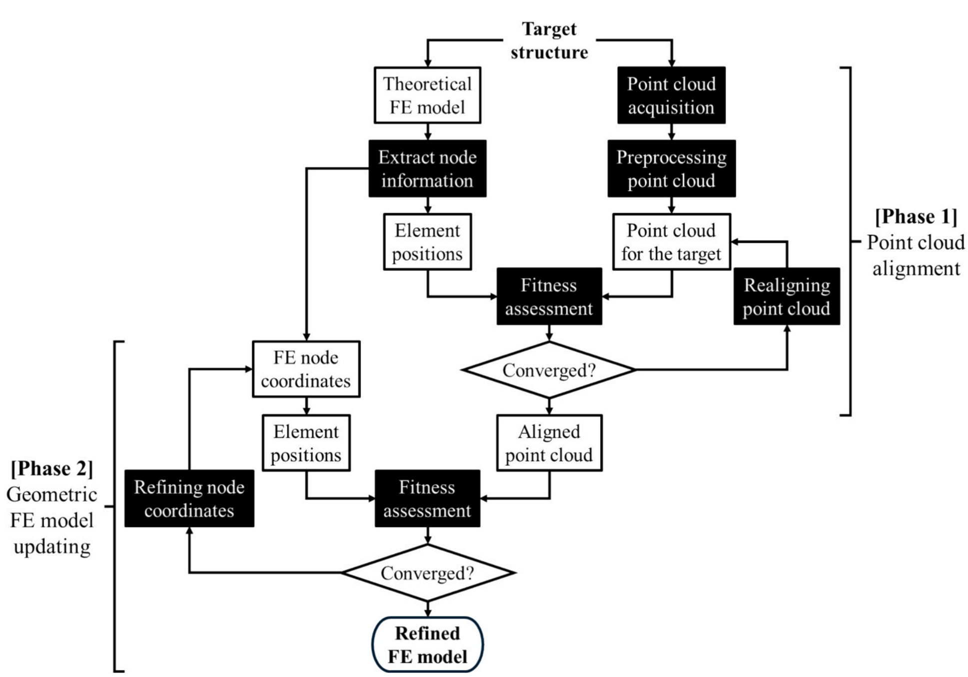

Fig. 1. Schematic description of the proposed method.

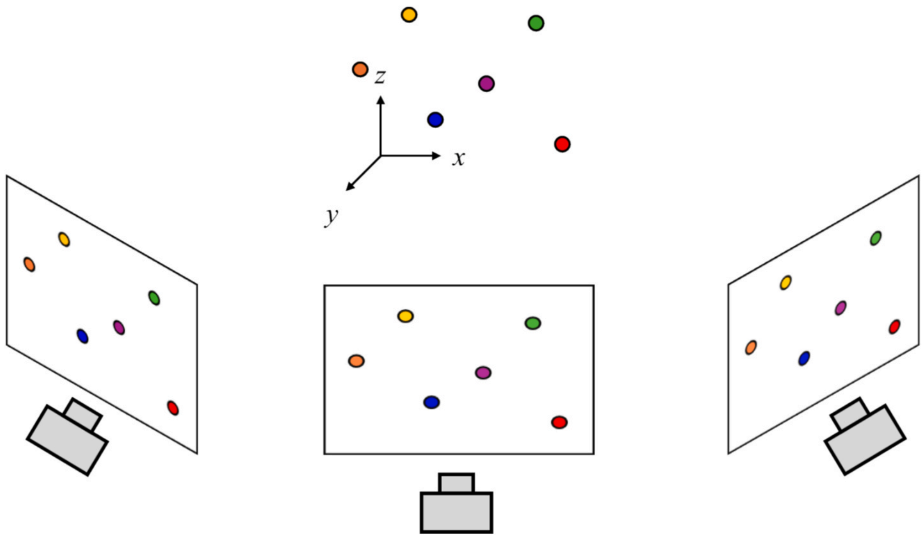

Fig. 2. Concept of Structure from Motion.

Feature point detection involves the process of identifying stable and common features that are robust to translation, rotation, and scale changes in each image. Principal feature point detection algorithms include SIFT (Scale-Invariant Feature Transform) [37] and SURF (Speeded Up Robust Features) [38], which are used to extract unique and distinctive feature points from images.

In the feature matching stage, descriptors are generated for the feature points in the images, and their similarities are compared to find the optimal match. However, since not all feature points can be matched precisely, techniques such as RANSAC (Random Sample Consensus) [39] are used to eliminate outliers and ensure the best possible matching.

As a result of feature matching, the matched feature points are used to extract the extrinsic parameters related to the camera position and orientation from each image. This step is essential for improving the accuracy of reconstructing the target object in 3D space by mapping the viewpoints extracted from each image.

After estimating the extrinsic parameters of the camera from each image, optimization is performed between the extracted camera parameters and the matched feature points for 3D reconstruction of the target object. During this optimization process, the feature points in the images are mapped to 3D space based on the projection matrix, finally resulting in the generation of a point cloud for the target object. The obtained point cloud can serve a vital role in providing fundamental data for various engineering and structural analysis tasks.

2.2. Preprocessing point cloud

Once the point cloud of the target structure is obtained, it is then used to match with the theoretical FE model and update the geometric characteristics of the model to align with those of the actual structure. Prior to this, appropriate preprocessing of the point cloud is required for computational efficiency. This may include removing unnecessary point clouds, applying noise filtering, and adjusting the density of the points.

The most common method for data cleaning of point clouds is region-based filtering. This method involves removing points from specific areas that are not relevant to the analysis, such as points from external regions like the ground or background. To achieve this, spatial boundaries are set, or a region of interest (ROI) is specified to eliminate points outside the target area.

Another method that can be applied for point cloud data cleaning is color-based filtering. This method filters the point cloud using color information when the target object or irrelevant background elements have distinct color ranges. When the target structure and background elements exhibit different color characteristics, points within the specified color range are retained or removed, allowing for the sorting of meaningful portions of the point cloud and thereby enhancing the accuracy of the data.

Statistical filtering and outlier removal methods can be applied for noise reduction in point clouds. Statistical filtering removes points that deviate from the statistical consistency with their neighboring points, based on the distance or density relative to nearby points. This approach typically distinguishes noise or external elements by local density differences. Outlier removal identifies and eliminates irregularly scattered points through cluster analysis. The outliers typically lie far from the geometric boundary of the structure, and they can be filtered out by setting a threshold distance.

2.3. Point cloud alignment

Before updating the geometric factors of the FE model, aligning its coordinate system with that of the point cloud is essential. The coordinate alignment of the point cloud involves iteratively adjusting the rotation, translation, and scaling of the point cloud to match the coordinates of the FE model. This process employs a scale factor and transformation matrix to convert all points in the point cloud into the coordinate system of the FE model, ensuring the best possible fit between the point cloud and the target FE model.

Let each point in the point cloud be denoted as $P = (x_p, y_p, z_p)$, and each element E of the FE model be defined as either a line element L or a shell element S. The shortest distance between a point in the point cloud and an element of the FE model is defined as shown in Fig. 3 and is calculated using Eq. (1).

$$d_{ij}(P_i, E_j) = \min_{X_j \in E} \|P_i - X_j\| \tag{1}$$

Here, P and E represent a point in the point cloud and an element of the FE model, respectively, and X denotes the set of points belonging to the element E of the FE model. The notation $||\cdot||$ represents the Euclidean distance between two points.

When the distance between a specific point in the point cloud and an element of the FE model is less than a certain threshold, the point is considered an inlier matching with the FE model (Fig. 4). The threshold can be determined based on the thickness or size of the element. The process is then iteratively repeated for all points by updating the scale factor and transformation matrix applied to the point cloud. The point cloud, in which as many points as possible are evaluated as inliers, is considered to be aligned with the coordinate system of the FE model. To

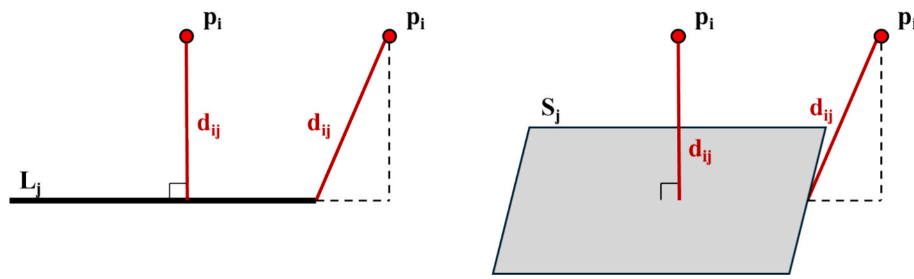

Fig. 3. Shortest distance between the point and the FE model elements.

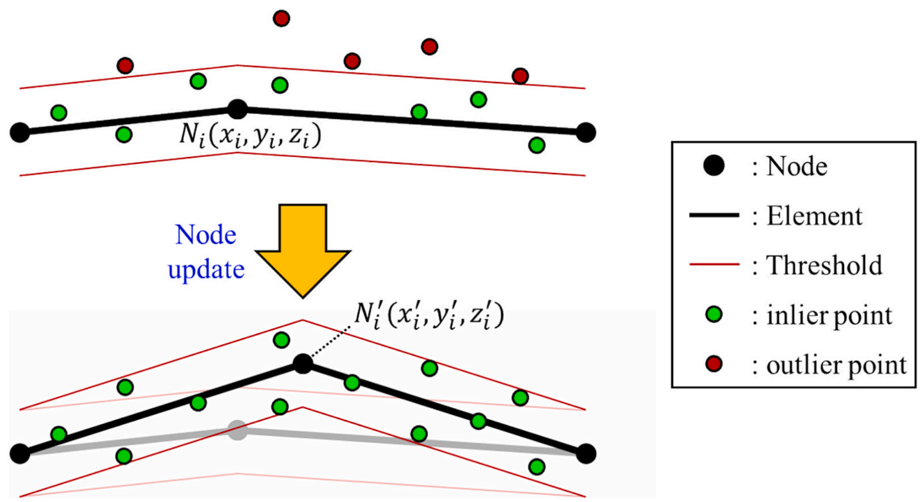

Fig. 4. Coordinate updating of the FE model based on distance to the point cloud.

achieve this, the optimization process, as shown in Eq. (2), is iteratively performed to adjust the scale factor (s) and transformation matrix (T) in a way that maximizes the number of inlier points. During this process, rotation, translation, and scaling transformations are applied to induce alignment between the FE model and the point cloud. It should be noted that if the scale of the point cloud is set too small, all points could be regarded as inliers. Therefore, appropriate constraints should be applied to the variables involved in transforming the point cloud based on engineering judgment.

$$\max_{s,T} \sum_{i=1}^{N} \sum_{j=1}^{M} \delta\left(d_{ij}\left(sT \bullet P_i, E_j\right) \leq \varepsilon\right) \tag{2}$$

In the above, s and T represent the scale factor and transformation matrix for the point cloud. P denotes the points in the point cloud, and E represents the elements of the FE model. The d_{ij} refers to the shortest distance between the point cloud and the FE model, while ε is the threshold used to define inlier points. δ is an indicator function that returns 1 when the condition is true and 0 when it is false. N and M represent the number of points in the point cloud and the number of elements in the FE model, respectively.

2.4. Geometric FE model updating

Once the coordinate system alignment between the theoretical FE model and the point cloud is completed, the next step involves refining the geometric features of the FE model based on the point cloud of the actual structure, which may not perfectly match the theoretical FE model. In the previous step, the variables for fitting the point cloud and the FE model were the scale of the point cloud and the transformation matrix. In contrast, in this step, the aligned point cloud is fixed, and the node coordinates of the FE model are fine-tuned to further enhance the geometric alignment between the FE model and the point cloud. The optimization is still performed based on the shortest distance between the point cloud and the FE model elements. The objective of the optimization is to ensure that the shortest distance between each point in the point cloud and the elements of the FE model is within the threshold value ε, and this optimization process, as shown in Eq. (3), is iteratively performed.

$$\max_{X} \sum_{i=1}^{N} \sum_{j=1}^{M} \delta\left(d_{ij}\left(P_i, E_j(X)\right) \leq \varepsilon\right) \tag{3}$$

Here, X represents the set of node coordinates of the FE model to be optimized. Through this process, the FE model is adjusted to more precisely align with the point cloud, which contains the 3D information of the actual structure, minimizing the geometric difference with the target.

3. Simulation test

The simulation test was conducted to analyze the effectiveness of the proposed method under controllable conditions with known solutions. In this stage, the theoretical reference model was used as the initial state of the FE model, and an arbitrarily manipulated model was set as the target. The test aimed to assess whether the proposed method could effectively update the geometry of the FE model as intended.

3.1. Simulation setup

The theoretical reference model for the simulation test is shown in Fig. 5. The model is a suspension bridge with a girder width of 240 mm and a length of 4000 mm. Rather than representing an actual structure, it adopts the same specifications as the model used in the experimental

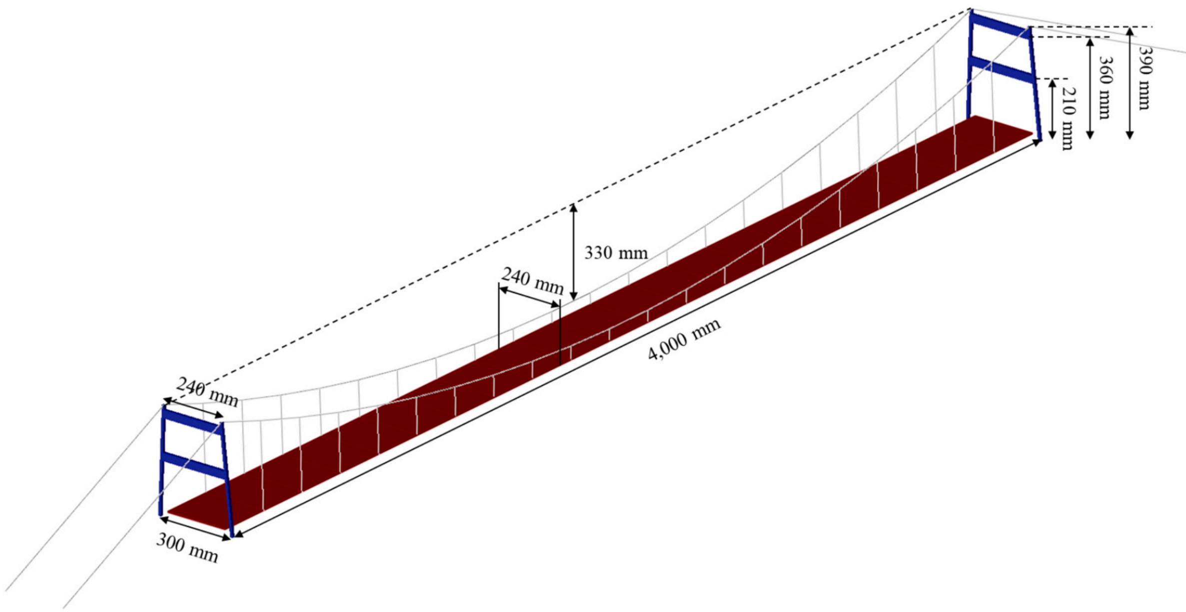

Fig. 5. Reference FE model.

test in the following chapter. The two main cables are supported by two pylons, with each cable carrying 20 suspenders on each side. The sag height, which plays a crucial role in the overall structural stability and load distribution of the suspension bridge, is set to 330 mm. The detailed sectional properties of each member are provided in Table 1.

The material properties used to determine the deflection shape of the model are applied as shown in Table 2. Fixed boundary conditions are imposed at the bottom of the pylons, while the girder is simply supported. In the simulation test, the reference model is a theoretical design model that does not consider fabrication and construction errors of the actual structure. It serves as the initial state for updating the geometric shape of the FE model.

In the simulation test, the target model for geometric updating is defined by applying arbitrary deformations to the reference model, as shown in Fig. 6. The characteristics of the target model are as follows: First, the sag height was adjusted from the original 330 mm to 297 mm, a reduction of 10 %. Second, the position (height) of the cross beam at each pylon was altered by 10 %. Third, the position of the suspenders attached to the main cables on each side was shifted by 10 %. Finally, dead load was applied to the manipulated model, and the deformed shape was used as the final target model for geometric updating of the FE model.

3.2. Point cloud generation

In this study, a simulation test was conducted to analyze the feasibility of the proposed point cloud-based geometric FE model updating method. To this end, the point cloud used here was not obtained through conventional methods but was instead manually generated to correspond to the geometric information of the target model for analysis.

Generally, a denser point cloud is more advantageous as it can provide more detailed geometric information about the target structure, which helps distinguish it from noise. The simulation test evaluates how effectively the known information serves its intended purpose. In this study, a point cloud was generated with approximately 0.01 points per cubic millimeter (mm^3) of the FE model, considering both geometric resolution and computational efficiency.

Table 1
Sectional properties of members of the FE model.

Member	Section type	Sectional properties
Girder	Plate	t = 4.5 mm
Main cable	Circular	A = 4.65 mm^2
Suspender	Circular	A = 0.33 mm^2
Pylon shaft	Rectangular	b x h = 10 x 10 mm
Pylon crossbeam	Rectangular	b x h = 10 x 30 mm

Table 2
Material properties of the FE model.

Elastic modulus	Density
210 Gpa	7,850 kg/m^3

Fig. 7 shows the point cloud generated for the target FE model, and the range of the point cloud generation was set based on the sectional properties of each member of the FE model. The total number of points in the generated point cloud is 14,502.

3.3. Point cloud alignment

In this section, the alignment of the point cloud is performed to match the coordinate system between the point cloud and the FE model. Since the point cloud generated in the simulation test was created based on the target FE model, it is initially aligned with the coordinate system of the FE model. Therefore, to analyze the effectiveness of the proposed point cloud alignment method based on the distance between the FE model and the point cloud, the initial state of the point cloud was arbitrarily manipulated, as shown in Table 3, to deviate from the coordinate system of the FE model.

To align the point cloud, an optimization process was performed by setting the scale factor of the point cloud and the rotation and translation along the x, y, and z directions as variables, with the objective function designed to maximize the number of points whose shortest distance from the FE model is less than the threshold. Various optimization algorithms can be employed in this process, considering the size of the variables, as well as the nonlinearity and multi-dimensionality of the objective function. In this study, a genetic algorithm [40], known for its strength in global optimization and efficiency in complex search spaces, was used.

Table 4 shows the parameters of the genetic algorithm used for the point cloud alignment. The population size represents the number of

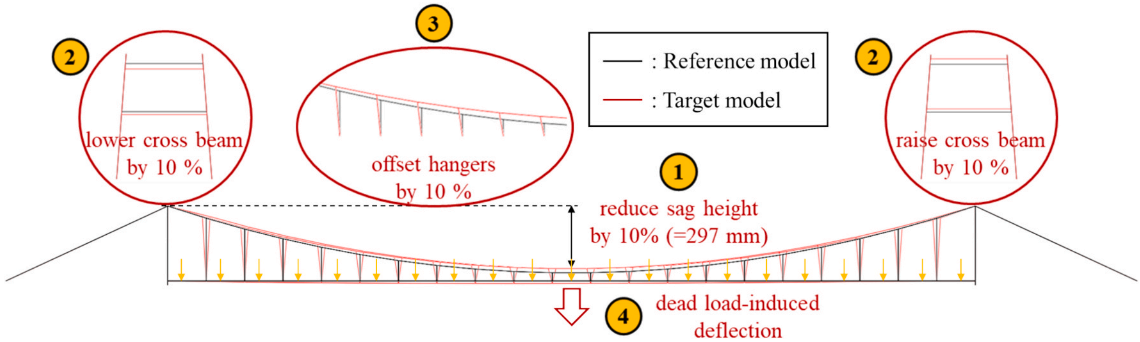

Fig. 6. Target model for the simulation test.

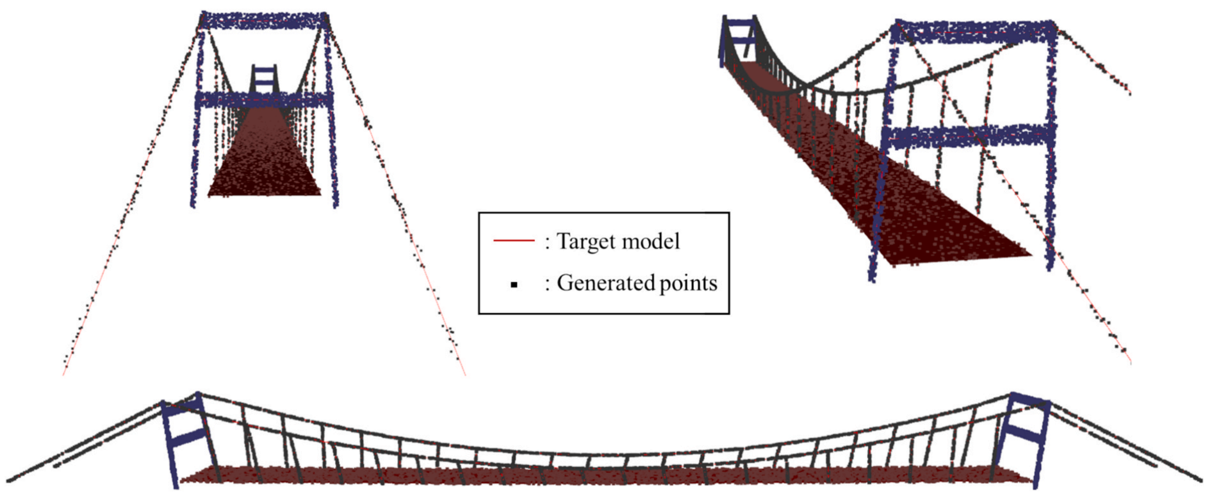

Fig. 7. Generated point cloud corresponding to the target model.

Table 3
Initial coordinate setup for the point cloud in the simulation test.

Scale factor	Rotation [deg]			Translation [mm]		
	x	y	z	x	y	z
0.97	1	−1	1	10	−10	10

Table 4
Genetic algorithm parameters for point cloud alignment.

Population size	Number of generations	Crossover rate	Mutation rate
50	200	0.8	0.1
100			
250			
500			

individuals considered in one iteration of the optimization process. As this number increases, the search for optimization becomes more comprehensive, but the computation time also increases. The number of generations refers to the number of iterations in which the optimization process is performed, and sufficient repetitions are necessary to ensure convergence. In a genetic algorithm, crossover refers to the process of combining two individuals to create a new one, while mutation involves the probabilistic modification of an individual to ensure diversity in the search space.

Fig. 8 shows the optimization process for point cloud alignment using a genetic algorithm. In this process, the criterion for determining whether a point is an inlier in the objective function is based on the shortest distance from the FE model, which is derived from the sectional properties of each member. In the genetic algorithm, which has 7 variables for optimization and a population size of 50, approximately 30.35 % (4401 points) of the total 14,502 points were identified as inliers. For larger populations, the performance of the genetic algorithm showed convergence. Finally, the alignment resulted in approximately 47.88 % (6943 points) of the points being identified as inliers. This suggests that, based on the point cloud information, the geometric characteristics of the theoretical model and target model are fundamentally aligned at around 47.88 %. The remaining 52.12 % of the points, which were not identified as inliers, represent areas where arbitrary deformations were applied to the theoretical model and provide information for the geometric updating of the target model.

The parameters for the aligned point cloud obtained through the optimization process are presented in Table 5. Here, x represents the longitudinal direction, y represents the transverse direction, and z represents the vertical direction. Among the considered parameters, the scale factor, which has the most significant impact on the point cloud alignment, showed a very accurate result with only a −0.02 % error when compared to the ground truth value. While relatively acceptable errors were observed in the x and z direction rotations and the x and y direction translations, significantly large errors were observed in the y direction rotation and particularly in the z direction translation.

The inlier identification results of the points at the initial state before the alignment and the final result after optimization and alignment are shown in Fig. 9. In the simulation test, when manipulating the target model from the reference model, the deflection of the girder was applied. Since the point cloud was generated with a certain number of points per unit volume for the target model, a large number of points were concentrated on the girder, which occupies a large volume

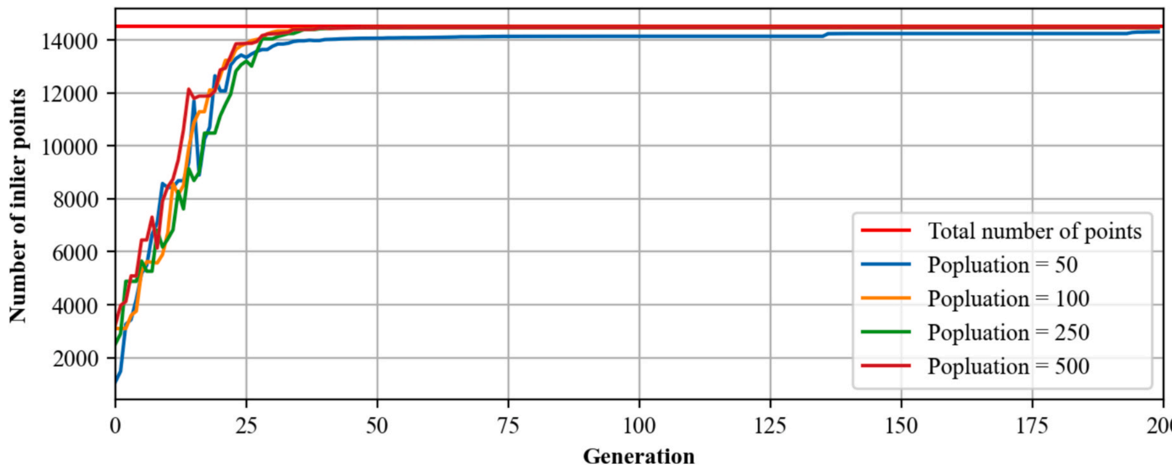

Fig. 8. Optimization process for point cloud alignment in the simulation test.

Table 5
Point cloud alignment parameters relative to the initial state in the simulation test.

	Scale factor	Rotation [deg]			Translation [mm]		
		x	y	z	x	y	z
Ground truth	1.0309	−1	1	−1	−10	10	−10
Optimizedresults	1.0264	−0.9286	0.9899	−1.0110	−8.2568	10.5255	−9.9261
Error	−0.44 %	−7.14 %	−1.01 %	1.10 %	−17.43 %	5.25 %	−0.74 %

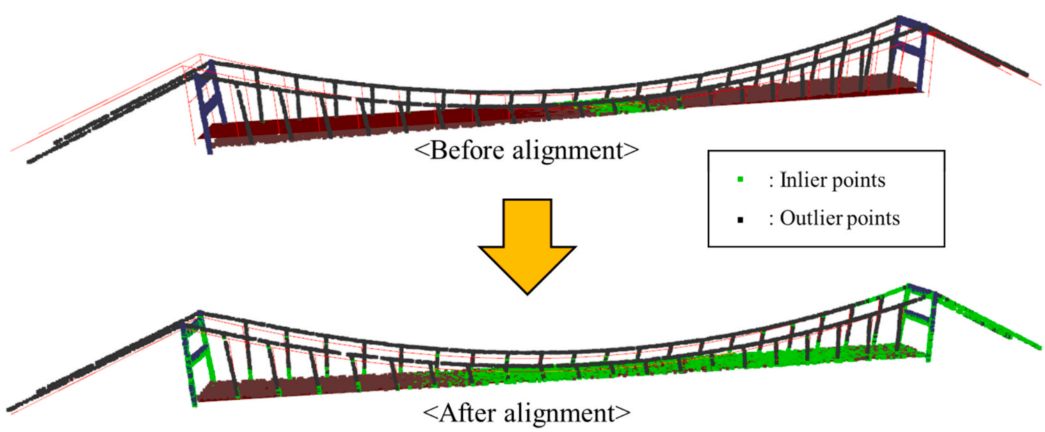

Fig. 9. Point cloud alignment result of the simulation test.

compared to other members. As a result, during the point cloud alignment optimization process based on the distance between the FE model and the point cloud, a significant error in the z-direction translation occurred in order to match the large number of points concentrated on the girder with the FE model. Additionally, due to the height difference between the girder and the main cables in the reference and target models, it was not possible to match the points corresponding to the girder and cables simultaneously. Consequently, an error occurred in the y-direction rotation in order to match at least one side of the cables.

The alignment of the point cloud is the step in which the basis for updating the geometric shape of the FE model is established. Since the final geometric characteristics of the FE model are updated based on this alignment, it is crucial for determining the geometric accuracy and consistency of the FE model. As shown in the simulation test results, geometric discrepancies between the theoretical model and the target model can prevent correct alignment. Therefore, appropriate post-processing is required in this process based on engineering judgment.

3.4. Geometric FE model updating

In this section, the geometry of the FE model is updated through an optimization process, using the point cloud of the target model as the reference and the node coordinates of the FE model as variables. The FE model used in this study was not designed for analyzing structural behavior but for utilizing its geometric characteristics. Therefore, its nodes were not subdivided to a level sufficient for accurately representing the structural behavior, and only the joint points of each member were used as nodes. The FE model consists of 108 nodes, and considering the coordinates in the x, y, and z directions, a total of 324 variables are considered in the optimization process for the geometric updating of the FE model.

This section also adopts a genetic algorithm for optimization, with the objective function set to maximize the number of points whose shortest distance between the point cloud and the FE model satisfies a threshold. However, while there are six variables for the alignment of the point cloud in the previous section, for the geometric refinement of the FE model, adjustments must be made for all nodes, resulting in a total of 324 variables. Therefore, the parameters applied to the genetic algorithm need to be expanded. In this study, various populations of the genetic algorithm were considered, as shown in Table 6, and the performance of the optimization process was analyzed accordingly.

Fig. 10 shows the optimization process for the geometric updating of the FE model. For all population conditions considered, the genetic algorithm converged within 100 generations. For the smallest population size of 1000, the number of points classified as inliers was 11,013, which accounted for only about 75.94 % of the total points. As the population size increased, the performance of the algorithm gradually improved. With a population size of 50,000, the number of points classified as inliers was 14,231, which accounted for about 98.13 % of the total points.

The performance of the proposed method was quantitatively analyzed based on the coefficient of determination (R^2) in Eq. (4). The R^2 value, the closer it is to 1, indicates that the updated model matches the target model more accurately.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{4}$$

Here, N represents the total number of node coordinates, y_i denotes the node coordinates of the target model, $\bar{y}$ is the mean value of the target model node coordinates, and $\hat{y}_i$ represents the node coordinates of the updated model from the initial state. The scatter plot, with the node coordinates of the target model on the x-axis and the node coordinates of the updated model on the y-axis, is shown in Fig. 11. As seen in Fig. 11(a) and (b), the coordinates in both the x and y directions for the initial and updated states yielded R^2 values exceeding 0.9999. This is because the node coordinates are widely distributed from the mean value, and the model, arbitrarily set as the target model in the simulation test, primarily applied changes in the z-direction from the initial state. For the z-direction coordinates, the R^2 value, initially 0.9919, was improved to exceed 0.9999, demonstrating that the geometric updating of the FE model based on the point cloud was effectively performed.

The performance of the proposed method, analyzed based on the mean distance between the nodes of the updated model and the target model, is shown in Fig. 12. While the mean node offset between the initial state (the theoretical reference model) and the target model was approximately 13.21 mm, the mean node offset of the updated model through the proposed method was only about 0.59 mm. This error is considered acceptable relative to the dimensions of the model, indicating that the proposed updating process has been effectively performed.

Details of the FE model before and after the geometric updating are shown in Fig. 13. Most of the nodes were well adjusted, as intended, to align with the target model based on the information of the point cloud. For more precise performance, more accurate point cloud extraction techniques and the application of more advanced optimization methods will be required.

Table 6
Genetic algorithm parameters for geometric updating of FE model.

Population size	Number of generations	Crossover rate	Mutation rate
1,000	200	0.8	0.1
2,000			
5,000			
10,000			
20,000			
50,000			

4. Experimental test

4.1. Experimental setup

An experimental test was conducted under laboratory conditions to assess the field applicability of the proposed method. For the test, a suspension bridge model with the same design parameters as those used in the simulation test was fabricated, and the proposed method was applied.

Fig. 14 shows an overview of the target model used in the experimental test. Although it was fabricated to have the same geometric characteristics as the theoretical reference model, it inevitably includes imperfections such as manufacturing or construction errors. In approaches to analyze the structure based on the FE model, even when the model is updated with the same materials and other properties as the target structure, these geometric imperfections can lead to inaccuracies in the analytical results.

4.2. Point cloud acquisition

In the experimental test, the acquisition of the point cloud for the target structure utilized the SfM technique, which involves using images captured from various angles of the target object with a single camera. This technique derives the point cloud of the target object and the camera poses for each image through correspondence search and incremental reconstruction processes. The SfM results, using a total of 57 images, are shown in Fig. 15.

The raw point cloud, consisting of 1,015,808 points, was obtained without any preprocessing. It included elements such as the surrounding background and unnecessary experimental equipment, which would later require unnecessary calculations for the geometric updating of the FE model based on the point cloud coordinates. Therefore, for computational efficiency, the obtained point cloud needs to be preprocessed using appropriate methods.

The most basic preprocessing step, which can be applied first, is an area-based filtering method to remove points from external regions, rather than from the target object. In this process, the user manually defines the ROI (Region of Interest) for the target object to eliminate points from the external regions. The application of area-based filtering is shown in Fig. 16, and as a result, the total number of points in the point cloud was reduced by approximately 93.85 %, leaving 62,466 points.

However, the point cloud still contained unnecessary elements, such as supporting structures, which were not part of the target structure. These elements were removed through manual cropping, and the final point cloud for applying the proposed method is shown in Fig. 17. This point cloud consists of 11,560 points, which represents a reduction of approximately 98.86 % compared to the raw state.

Additional preprocessing methods that could be applied include statistical filtering based on the distance and density of neighboring points, or the removal of external points based on clustering. However, due to the characteristics of the suspension bridge model used in this study, where points may not be densely formed on thin components such as cables, additional preprocessing steps were not applied, based on engineering judgment, to prevent the removal of such points through outlier filtering. Such preprocessing steps can be further applied, taking into account the characteristics of the structure and the formed point cloud.

4.3. Point cloud alignment

The initial state of the preprocessed point cloud was manually aligned, and an optimization process was performed for coordinate alignment using a genetic algorithm. The variables for the point cloud alignment optimization process include the scale factor, and rotations and translations in the x, y, and z directions. The objective function is to

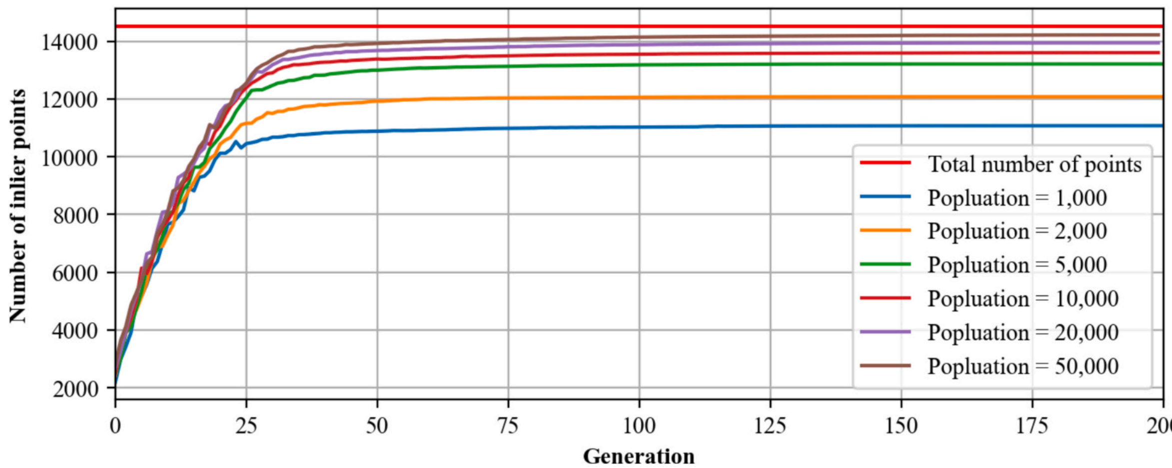

Fig. 10. Optimization process for geometric updating of FE model in the simulation test.

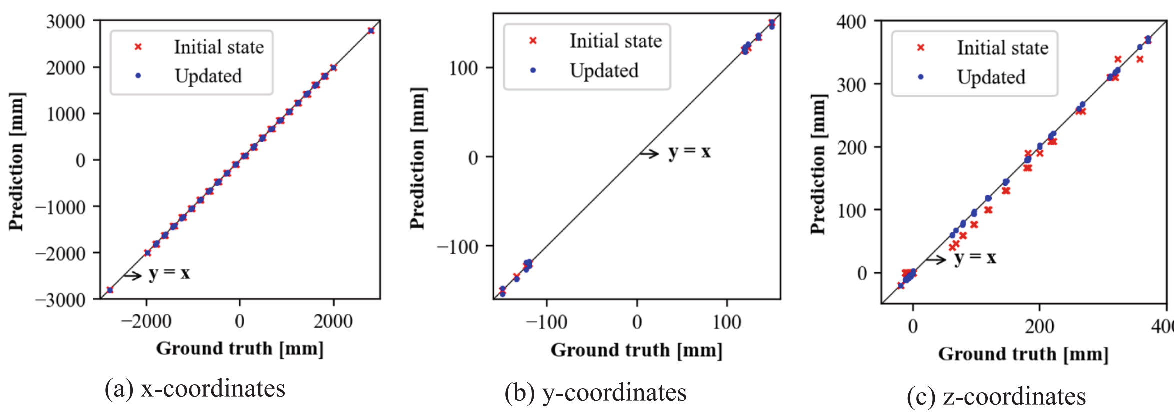

Fig. 11. Comparison of node coordinates following geometric model updating.

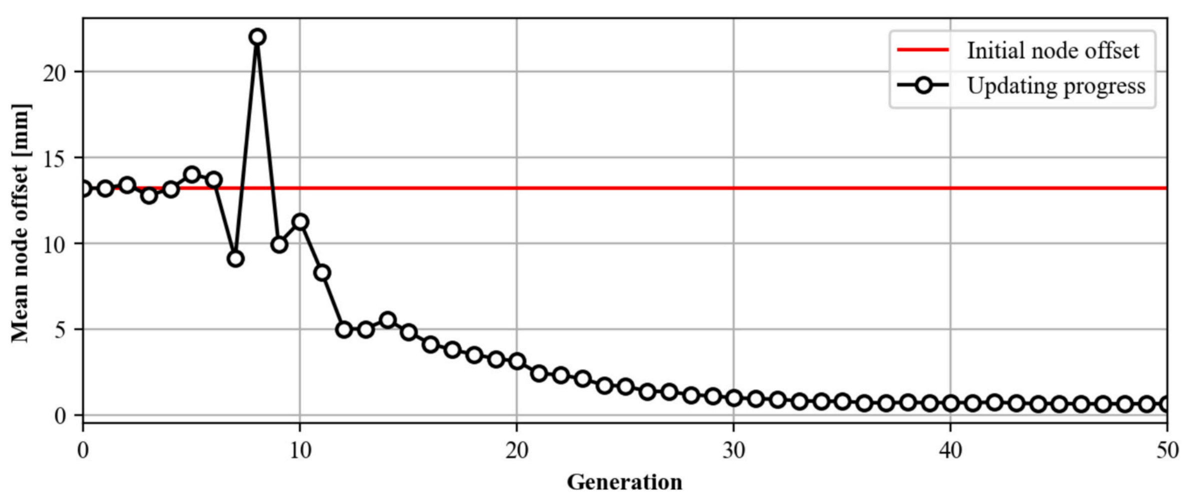

Fig. 12. Mean node offset relative to the target model during the optimization process.

maximize the number of points where the shortest distance from the FE model satisfies the dimensions of each component. This process is shown in Fig. 18, and it demonstrated that the point cloud alignment converged sufficiently within the considered iteration count. The parameters related to the aligned point cloud, compared to the initial state, are provided in Table 7.

The number of points identified as inliers in the manually set initial state of the point cloud was 1723, and through alignment, a coordinate system was derived in which a total of 2838 points could be identified as inliers. The point cloud alignment in the experimental test is shown in Fig. 19.

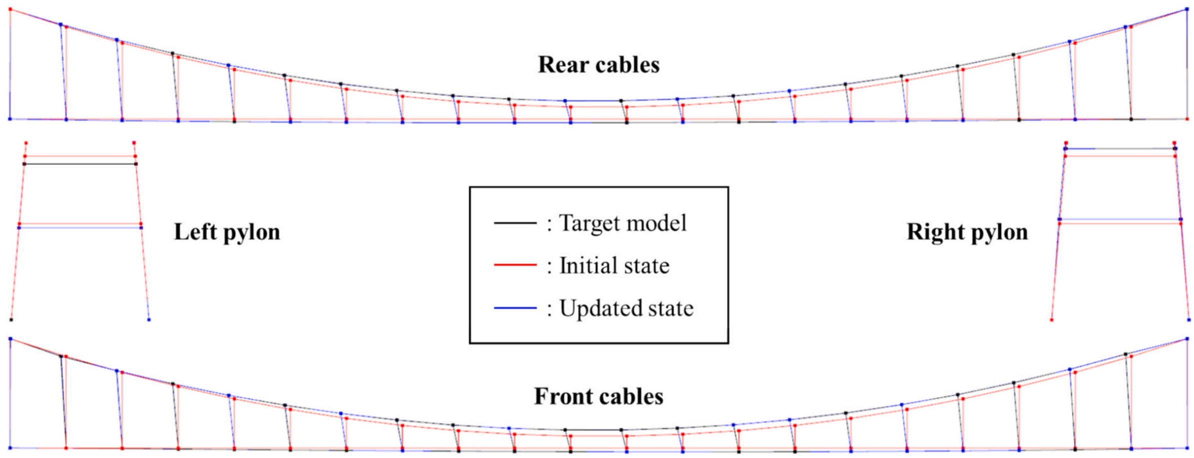

Fig. 13. Comparison of FE models before and after geometric updating in the simulation test.

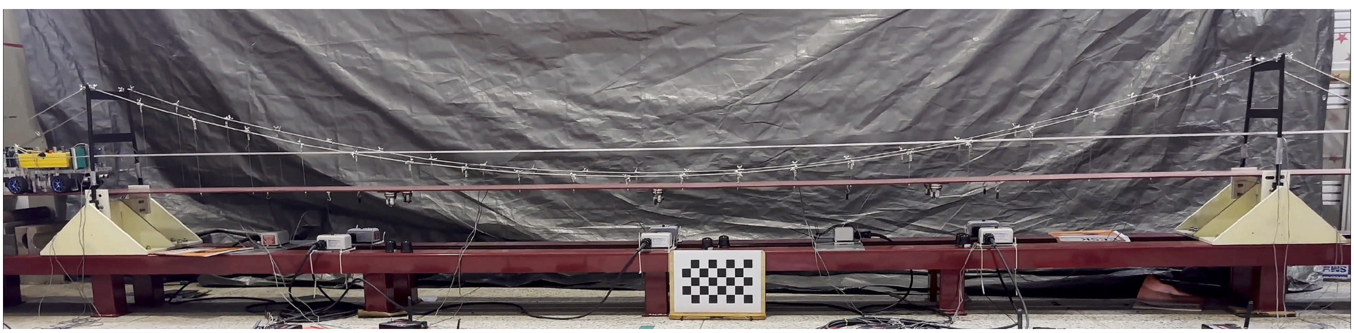

Fig. 14. Experimental setup.

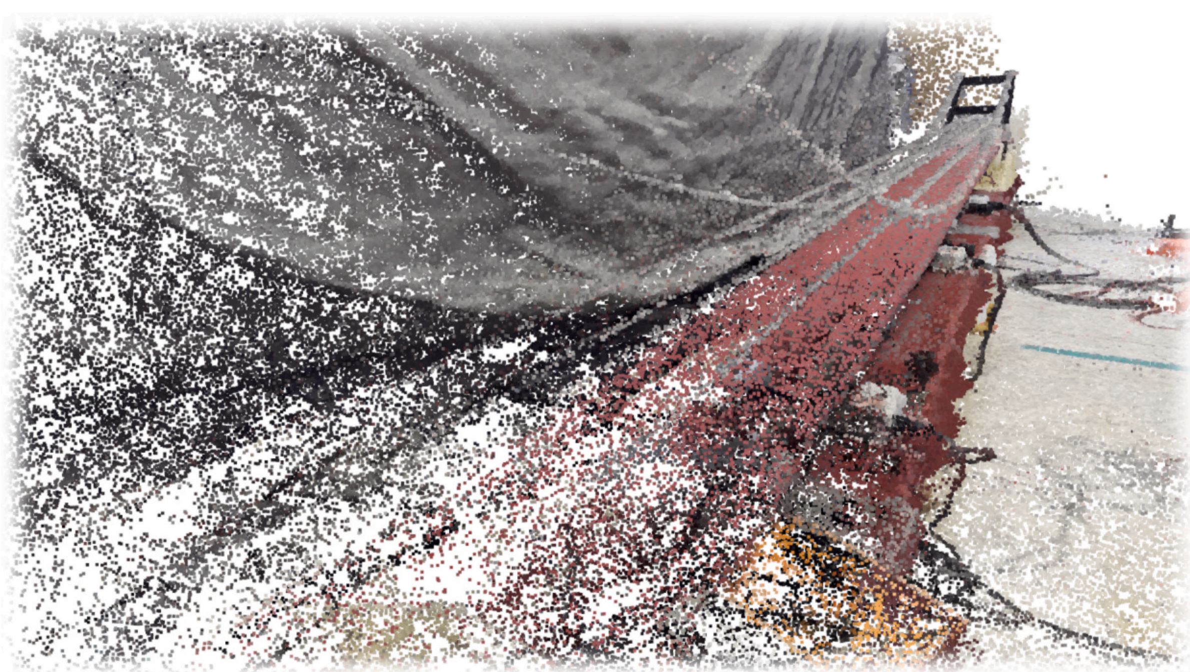

Fig. 15. Raw point cloud extracted by SfM.

4.4. Geometric FE model updating

Through the previous section, the coordinate system alignment between the point cloud of the target structure and the theoretical FE model was performed. In the aligned point cloud, points that did not satisfy a certain threshold for the shortest distance from the theoretical FE model and were not identified as inliers could be considered either noise or geometric information of the target structure for updating the

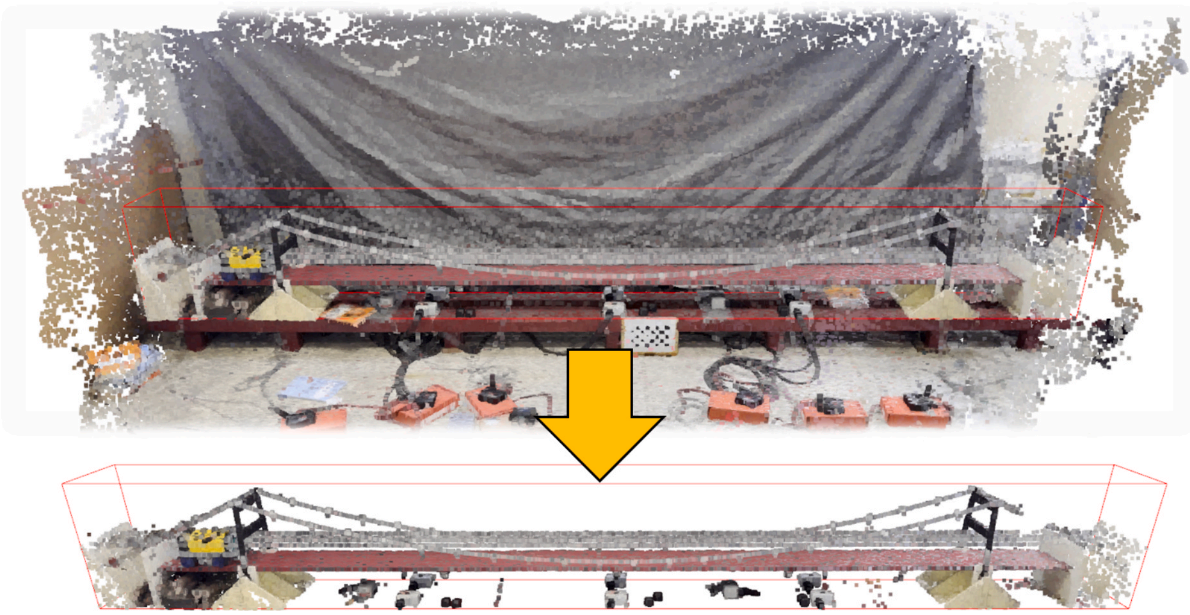

Fig. 16. Region-based filtering applied to the point cloud.

Fig. 17. Preprocessed point cloud for applying the proposed method.

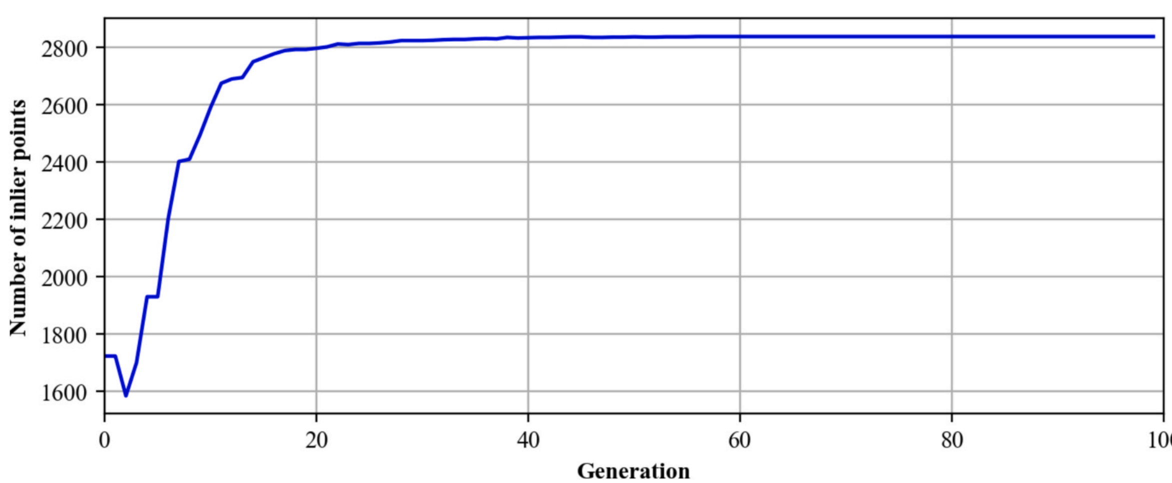

Fig. 18. Optimization process for point cloud alignment in the experimental test.

Table 7
Point cloud alignment parameters relative to the initial state in the experimental test.

Scale factor	Rotation [deg]			Translation [mm]		
	x	y	z	x	y	z
0.9996	−0.1982	−0.0014	−0.1630	6.1470	−4.2196	−2.3617

FE model. This study assumes that the point cloud of the target object is more densely formed compared to surrounding noise. Based on this assumption, the point cloud containing geometric information of the target structure, which is densely formed, has a greater influence in the proposed method than the relatively sparse noise. Consequently, the optimization process enables the refinement of the FE model so that it

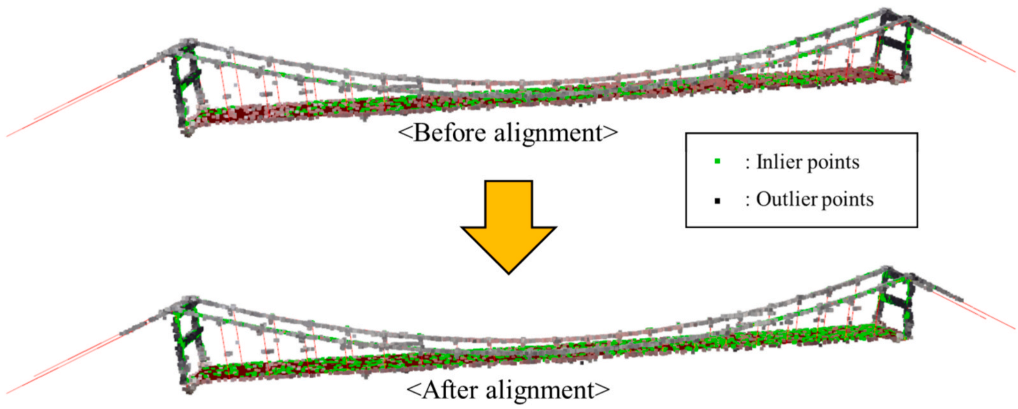

Fig. 19. Point cloud alignment result of the experimental test.

fits the geometric characteristics of the target structure.

The geometric updating process, which considers the node coordinates of the FE model as variables, is illustrated in Fig. 20. The initial FE model, which had 2838 inliers with respect to the target point cloud, was updated to the model with 3987 inliers through the proposed method. Details of the differences in the FE model due to the geometric updating are shown in Fig. 21. Since the proposed method was applied to a model designed for laboratory testing, the geometric differences were not represented significantly. However, for actual structures where fabrication and construction errors may be more pronounced, especially for aging structures, the differences from the theoretical model at the design stage can be even more significant.

5. Conclusion

In this study, a method for updating the FE model to account for the geometric imperfections of the target structure was proposed using a computer vision-based point cloud. The effectiveness of the proposed method was analyzed through simulation tests, and its applicability to real-world conditions was evaluated through experimental tests. Through the study, the following conclusions were drawn.

I. The geometric characteristics of the FE model can be updated based on computer vision by fitting the point cloud with the FE model, and the consistency can be evaluated based on the distances between the elements of the FE model and the points in the point cloud.
II. The geometric updating of the FE model can be performed through the alignment of the coordinate system and the refinement of the node coordinates. Simulation test results showed that the R^2 value between the node coordinates of the updated and the target model was above 0.9999, proving that the updated model accurately reflects the geometric characteristics of the target model.
III. Point clouds obtained under actual conditions may inevitably contain noise; however, if the noise is relatively sparse compared to the point cloud of the target object, the proposed method can be effectively applied. Furthermore, as point cloud extraction and preprocessing technologies continue to advance, the effectiveness of the proposed method can be further enhanced.

The proposed geometric updating method for the FE model corrects errors that may occur during the construction and fabrication stages, reducing the differences between the theoretical design state of the FE model and the actual state of the structure. This not only enables more effective finite element model updating, improving the accuracy of the FE model and enhancing the reliability of simulations, but also allows for more accurate identification of deformations or damage in the structure, thereby enabling a more rational assessment of the structural safety.

While the feasibility of the proposed method was validated through simulation tests and its practical applicability was demonstrated through experimental implementation, it currently represents a proof-of-concept stage in which the process consists of relatively basic operations. As such, several limitations remain to be addressed. First, this

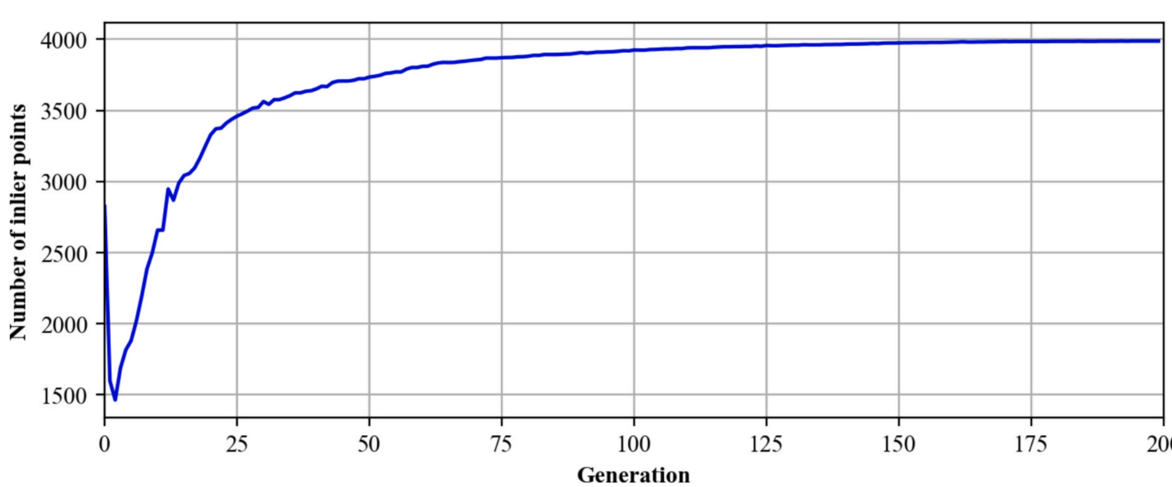

Fig. 20. Optimization process for geometric updating of FE model in the experimental test.

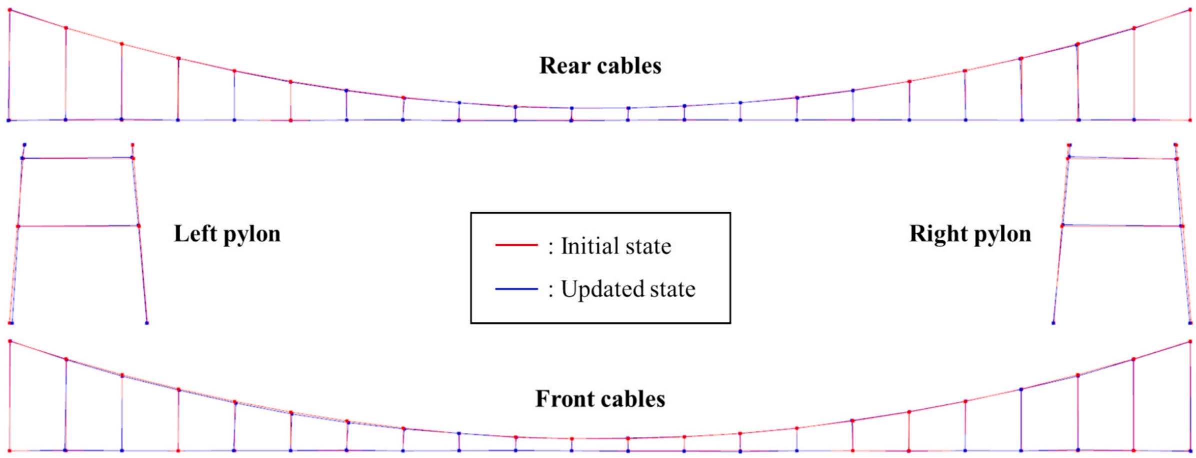

Fig. 21. Comparison of FE models before and after geometric updating in the experimental test.

study did not examine the influence of input image quality, which may affect the performance of the method in real-world applications. Furthermore, to enhance the robustness and automation of the process, future research may incorporate more advanced techniques such as learning-based feature extraction and semantic segmentation. Additional studies should also evaluate the computational efficiency of the overall framework and assess its effectiveness under a broader range of real-world scenarios.

CRediT authorship contribution statement

Yunwoo Lee: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jae Hyuk Lee:** Writing – review & editing, Validation, Investigation, Formal analysis. **Namju Byun:** Writing – review & editing, Validation, Formal analysis, Data curation. **Hyungchul Yoon:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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